

JEE MAIN REVISION MASTER: PROBABILITY SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 232 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Probability - Core Application	232 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Probability - Core Application (Very High Yield — 232 PYQs)

Core Principle: Key problem pattern in Probability.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A candidate has to go to the examination centre to appear in an examination. The candidate uses only one means of transportation for the ent..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "A letter is known to have arrived by post either from KANPUR or from ANANTPUR. On the envelope just two consecutive letters AN are visible. ..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "From a lot containing 10 defective and 90 non-defective bulbs, 8 bulbs are selected one by one with replacement. Then the probability of get..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

A letter is known to have arrived by post either from KANPUR or from ANANTPUR. On the envelope just two consecutive letters AN are visible. The probability, that the letter came from ANANTPUR, is:

- (A) $(7 / 10)$
- (B) $(10 / 17)$ **[Correct]**
- (C) $(12 / 19)$
- (D) $(7 / 19)$

Correct Answer: B

Solution Analysis: Let A = event that the letter came from ANANTPUR K = event that the letter came from KANPUR B = event that on the envelope, the two visible consecutive letters are AN We have to find: $P(A|B)$ Using Bayes' theorem, $P(A|B) = \frac{P(A)P(B|A)}{P(A)P(B|A) + P(K)P(B|K)}$ Since the letter is known to have come either from ANANTPUR or KANPUR, and no other information is given, we take $P(A) = P(K) = (1 / 2)$ So, $P(A|B) = \frac{P(B|A)}{P(B|A) + P(B|K)}$ Now we find these probabilities. 1. Probability of seeing AN in ANANTPUR Word: ANANTPUR Its consecutive pairs are: AN, NA, AN, NT, TP, PU, UR There are \$